Timeline Check-in

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1. Set up

Packages

```
# general use
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```
library(janitor)
```

```
Attaching package: 'janitor'

The following objects are masked from 'package:stats':

    chisq.test, fisher.test
```

```
# file organization
library(here)
```

```
here() starts at /Users/maisieprestridge/Desktop/github/group-project-193DD
```

```
# Boxplot Visualization

library(hrbrthemes)

library(viridis)
```

```
Loading required package: viridisLite
```

```
# text label instead of legend
library(cowplot)
```

```
Attaching package: 'cowplot'

The following object is masked from 'package:lubridate':

    stamp
```

```
# Ridgeline Plot
#install.packages("ggridges")
library(ggridges)
#install.packages("ggplot2")
library(ggplot2)

# Color Palette
library(NatParksPalettes)
library(wesanderson)
```

```
Registered S3 method overwritten by 'wesanderson':
  method        from
  print.palette NatParksPalettes
```

```r
# Use ragg_png device for better font rendering and system font support
knitr::opts_chunk$set(dev = "ragg_png")
```

Data

```r
# weather data from NOAA weather station
weather <- read_csv(here("data", "NOAA-weather-data.csv"))
```

```
Rows: 1912 Columns: 9
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (3): STATION, NAME, DATE
dbl (6): LATITUDE, LONGITUDE, ELEVATION, PRCP, TMAX, TMIN

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# water quality survey metadata
metadata <- read_csv(here("data", "NCOS_YSI_Water_Quality_Monitoring_0.csv"))
```

```
Rows: 687 Columns: 17
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (12): GlobalID, Monitoring Date, Monitor Name, Weather, Site Name, Othe...
dbl   (4): ObjectID, Water Surface Elevation, x, y
time  (1): Time of WQ Measurement

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# water quality parameter measurements
water_parameters <- read_csv(here("data", "YSI_Data_Begin_1.csv"))
```

```
Rows: 1488 Columns: 13
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (6): GlobalID, ParentGlobalID, CreationDate, Creator, EditDate, Editor
dbl (7): ObjectID, Depth Code, DO (mg/L), DO (%sat), Conductivity (uS/cm), S...
```

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Cleaning Code

Cleaning Weather

```
weather_clean <- weather |>
  # made column names all lowercase
  clean_names() |>
  # puts date in proper format and makes sure its understood as date (not as chr)
  mutate(date = mdy(date)) |>
  # keeping only these 4 columns of interest
  select(date, prcp, tmax, tmin) |>
  # creating new columns for month and year
  # extracting month from the date (needs to be in the standard date format for this functio
  mutate(month = month(date),
         year = year(date)) |>
  mutate(wy = case_when(
    # when month os october or later, assign a value of year +1
    # in the water ywar for next year
    month >= 10 ~ year + 1,
    # keeping all months < 10 the same (sept or earlier), assigning value of existing year
    .default = year
    ))
```

Cleaning Metadata

```
metadata_clean <- metadata |>
  # col names easier to work with (all lowercase w/ _)
  clean_names() |>
  # "!" is the "not-operator", columns we DONT want to keep, excluding
  select(!c(object_id, creator, editor)) |>
  mutate(monitoring_date = mdy_hms(monitoring_date)) |>
  rename(monitoring_datetime = monitoring_date) |>
  mutate(month = month(monitoring_datetime),
         year = year(monitoring_datetime)) |>
  # assigning water year
```

```
  mutate(wy = case_when(
    month >= 10 ~ year +1,
    .default = year )) |>
  # creatig new column for site full name
  mutate(site_full = case_when(
    site_name == "east_channel" ~ "East Channel",
    site_name == "phelps_bridge" ~ "Phelps Bridge",
    site_name == "venoco_bridge" ~ "Venoco Bridge",
    site_name == "other" ~ "Other"
  )) |>
  # filter out Other observations
  filter(site_full != "Other")
```

Cleaning Water Parameters

```
# creating a new object from water parameters
water_parameters_clean <- water_parameters |>
  # cleaning column names
  clean_names() |>
  # "!" not operator, excluding unneeded columns
  select(!c(creator, editor)) |>
  # renaming depth code to be elevation code to better represent methodology
  rename(elevation_code = depth_code) |>
  ## Everything below came from visualizing the data first and defining what to fix/add
  # fixing a mistake in the data
  mutate(elevation_code = case_when(
    # anytime theres a 10 in elevation code, replace it with a 1
    elevation_code == 10 ~ 1,
    # everything else, keep the same
    .default = elevation_code)) |>
  mutate(
    # turning column into a factor
    elevation_code = as_factor(elevation_code),
    # defining the correct order for the levels of the factor
    elevation_code = fct_relevel(elevation_code,
                                 # write in the order you want
                                 "0", "1", "2", "3", "4", "5", "6"))
```

Joining Code

Water Quality = Metadata + Water Parameters

```
# full joining
water_quality <- full_join(
  # left data frame
  x = metadata_clean,
  # right data frame
  y = water_parameters_clean,
  # name the columns in each df that you want to join by (naming the shared column)
  by = c("global_id" = "parent_global_id")) |>
  select(
    # information about the observation
    global_id, monitoring_datetime, time_of_wq_measurement, monitor_name, weather, site_name
    # water information
    water_surface_elevation, flow, notes,
    #measurement information
    elevation_code, do_mg_l, do_percent_sat, conductivity_u_s_cm, salinity_ppt, temperature_
  ) |>
  # taking the date out of this column, dont want the time
  mutate(date = date(monitoring_datetime)) |>
  # moving the new date column to a better location within the df
  relocate(date, .after = monitoring_datetime) |>
  # dont need this column anymore
  select(!monitoring_datetime)
```

Water Quality Weather = Water Quality + Weather

```
# left join
water_quality_weather <- left_join(
  # left data frame
  x = water_quality,
  # right data frame
  y = weather_clean,
  # joining these dfs by their date column
  by = "date")
```

Wrangling/Summarizing Code

Salinity Outliers

```
# creating new obj
salinity_outlier_ids <- water_quality_weather |>
  # filtering to only include outlier values
  filter(salinity_ppt > 1410) |>
  # extracting the global id column as a vector usign pull
  pull(global_id)
```

Summarizing Precipitation

```
# create new object from `water_quality_weather`
monthly_prcp <- water_quality_weather |>
  # only include elevation codes 0, 1, and 2
  filter(elevation_code %in% c("0", "1", "2")) |>
  # get rid of site names w/ "other"
  filter(!site_full == "NA") |>
  # removing any duplicate rows based on the combination of month, year, and wy
  distinct(month, year, prcp) |>
  # grouping by month and year
  group_by(month, year) |>
  # calculating the sum of prcp per month (not per site as it is a weather event and not site
  summarise(sum_prcp = sum(prcp, na.rm = TRUE)) |>
  # ungrouping
  ungroup() |>
  # creating a date column to be read as a date
  mutate(date = lubridate::make_date(.data$year, .data$month, 1)) |>
  # creating levels to `month`, putting them in a logical order
  mutate(month = factor(month,
                        levels = 1:12,
                        labels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "S
                        ordered = TRUE))
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by month and year.
i Output is grouped by month.
i Use `summarise(.groups = "drop_last")` to silence this message.
```

```
i Use `summarise(.by = c(month, year))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

Filtering and Summarizing Salinity

```
salinity_filtered <- water_quality_weather |>
  # keeping only elevation codes 0, 1, and 2
  filter(elevation_code %in% c("0", "1", "2")) |>
  # filtering the column global_id to exclude the values that match those in `salinity_outlie
  filter(!global_id %in% salinity_outlier_ids) |>
  # dropping the column for other_site_names
  select(!other_site_name) |>
  # get rid of site names w/ "other"
  filter(!site_name == "other") |>
  # setting the order of the levels (from lowest tp highest avg salinity value)
  # helps the order of the visualizations
  mutate(site_full = fct_relevel(site_full,
                                 "Phelps Bridge", "East Channel", "Venoco Bridge")) |>
  # taking out sites w/ NA
  filter(!site_full == "NA") |>
  # keeping only columns of interest (to reduce confusion)
  select(site_full, water_surface_elevation, elevation_code, salinity_ppt, temperature_c, mo
  # creating levels to `month`, putting them in a logical order
  mutate(month = factor(month,
                        levels = 1:12,
                        labels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "
                        ordered = TRUE)) |>
  # group by site full, month, and year
  group_by(site_full, month, year) |>
  # calculate mean salinity (per site per month))
  summarize(mean_salinity = mean(salinity_ppt, na.rm = TRUE),
            .groups = "drop")
```

Joining Monthly_prcp and Salinity_filtered

```
salinity_prcp <- salinity_filtered |>
  left_join(monthly_prcp, by = c("month", "year"))
```

Exploratory Figure Code

Exploring Salinity

```
# create new object `exploratory_salinity` from `water_quality_weather`
exploratory_salinity <- water_quality_weather |>
  # only keep elevation codes 0, 1, 2, and 3 (keeping three just for visualization)
  filter(elevation_code %in% c("0", "1", "2", "3")) |>
  # filtering the column global_id to exclude the values that match those in `salinity_outlie
  filter(!global_id %in% salinity_outlier_ids) |>
  # dropping the column `other_site_names`
  select(!other_site_name) |>
  # filterign out site names that match "other"
  filter(!site_name == "other") |>
  # setting the order of the levels (from lowest tp highest avg salinity value)
  # helps the order of the visualizations
  mutate(site_full = fct_relevel(site_full,
                                 "Phelps Bridge", "East Channel", "Venoco Bridge")) |>
  # taking out sites w/ NA
  filter(!site_full == "NA") |>
  # keeping only columns of interest (to reduce confusion)
  select(site_full, water_surface_elevation, elevation_code, salinity_ppt, temperature_c, mo
  # group by site, elevation code, month, and year
  group_by(site_full, elevation_code, month, year) |>
  # calculate mean salinity
  summarize(mean_salinity = mean(salinity_ppt, na.rm = TRUE),
            .groups = "drop") |>
  # creating levels to `month`, putting them in a logical order
  mutate(month = factor(month,
                        levels = 1:12,
                        labels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "
                        ordered = TRUE))
```

Exploring Water Surface Elevation

```
# create a new object `monthly_surface_elv` from `water_quality_weather`
monthly_surfcae_elv <- water_quality_weather |>
  # only include elevation codes 0, 1, and 2
  filter(elevation_code %in% c("0", "1", "2")) |>
  # Filter out site names matchign to "other"
```

```
  filter(!site_name == "other") |>
  # group by site, month, and year
  group_by(site_full, month, year) |>
  # calculate mean water surface elevation
  summarise(avg_elv = mean(water_surface_elevation, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # create a date column to be read as a date
  mutate(date = lubridate::make_date(.data$year, .data$month, 1)) |>
  # creating levels to `month`, putting them in a logical order
  mutate(month = factor(month,
                        levels = 1:12,
                        labels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "S
                        ordered = TRUE))
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by site_full, month, and year.
i Output is grouped by site_full and month.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(site_full, month, year))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
## Joining data frames
elv_viz <- monthly_surfcae_elv |>
  # joining by columns month, year, and site
  left_join(salinity_prcp,  by = c("month", "year", "site_full")) |>
  # filtering out all NA average elevation values
  filter(!avg_elv == "NaN")
```

Creating Precipitation Peaks

```
# creating a new object `precipitation_peaks` from `salinity_prcp`
precipitation_peaks <- salinity_prcp |>
  # including the dates of the top 4 monthly precipitation
  filter(date %in% c("2021-01-01", "2023-03-01", "2023-02-01", "2021-10-01")) |>
  # ungrouping data frame
  ungroup() |>
  # keeping these 4 distinct dates
  distinct(date, .keep_all = TRUE) |>
```

```
  # adding a label to each event
  mutate(label = c("Rain Event (5.19 in)", "Rain Event (2.06 in)", "Rain Event (1.80 in)", "
         # setting the y position for each event (dropping the 1.80 event lower for better v
         label_y = c(4.5, 4.5, 3.5, 4.5))

precipitation_peaks_facet <- precipitation_peaks |>
  cross_join(tibble(site_full = unique(salinity_prcp$site_full))) |>
  mutate(label_y_2 = rep(c(40, 45, 40, 50), times = 3))
```

\\

2. Ideas for analysis and background

The overarching question for our project is: How does precipitation explain salinity?

We expect that more precipitation will generally lead to lower salinity because rainfall adds freshwater and dilutes saltier water. However, salinity is also affected by water depth, evaporation and whether the ecosystem is connected to the ocean.

This is especially relevant as North Campus Open Space is a connected coastal wetland system. Because of this, the salinity is influenced by more than rainfall alone. Harrington and Wood (2014), although focused on an Australian wetland, help support this idea by showing that ocean-driven saltwater movement can also affect salinity. Rainfall can lower salinity by adding freshwater, while sea-level changes and saltwater entering the system can raise salinity again.

Additionally, between 2015 to 2016 salinity was reported to increase at NCOS during dry periods because evaporation concentrates salts, while rainfall can lower salinity by adding freshwater (Clark, 2016). Rainfall lowered salinity by adding freshwater to the slough, but major storm and breach events reconnected the slough to the ocean, allowing seawater to enter and raise salinity again (Clark, 2016). The report also notes that salinity often increases with depth, so precipitation helps explain salinity but is not the only factor (Clark, 2016).

References:

Clark, R. (2016). Water Quality of North Campus Open Space & Devereux Slough: Fall 2015 - Spring 2016. [online] Available at: https://escholarship.org/uc/item/2923f039 [Accessed 20 May 2026].

Wood, C. and Harrington, G.A. (2014). Influence of Seasonal Variations in Sea Level on the Salinity Regime of a Coastal Groundwater–Fed Wetland. Groundwater, 53(1), pp.90–98. doi:10.1111/gwat.12168.

Ideas for analyses to answer question and some preliminary code to run those analyses:

To answer our question, "How does precipitation explain salinity?", we analysed whether monthly precipitation is related to monthly average salinity across NCOS sites. Our draft code already cleans and joins the weather, metadata, and water-quality datasets, then calculates total monthly precipitation and mean monthly salinity by site.

We also chose to use Spearman's correlation coefficient to test whether salinity tends to decrease as precipitation increases.

The test gave a coefficient value of 0.0166 and p-value of 0.8251, suggesting that precipitation alone does not strongly explain salinity.

\\ # 3. Visualizations directly relevant to answering questions

Figures

Figure 1. - Monthly Precipitation at NCOS (Time Series w/ Rain Event Lines)

```
# base layer: ggplot
# data: salinity_prcp
ggplot(data = salinity_prcp,
       # x axis: date
       mapping = aes(x = date,
                     # y axis: monthly precipitation sum
                     y = sum_prcp)) +
  # prcp displayed by columns (a suggestion made by An after work plan/proposal edits)
  geom_col(color = "darkblue") +
  # adding themeing
  theme_bw() +
  # x-axis breaks every 6 months (to reduce cognitive load)
  # Format date axis labels as abbreviated month and full year
  scale_x_date(date_breaks = "6 months", date_labels = "%b %Y") +
  # angling labels to 45 degrees and to be right-aligned
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  # expanding limits to make room for text
  scale_y_continuous(limits = c(0, 6)) +
  # labeling x and y axes
  labs(x = "Date",
       y = "Total Monthly Precipitation",
       # adding title
       title = "Total Monthly Precipitation at NCOS") +
  # removing legend, redundant
```

```
  theme(legend.position = "none") +
  # adding in lines for prcp events
  geom_vline(data = precipitation_peaks,
             # centering them on dates
             aes(xintercept = date,
                 # coloring them by their established label
                 color = label),
             # signified by dotted lines
             linetype = "dotted",
             # width of lines
             linewidth = 1.25) +
  # adding custom color palette
  scale_color_manual(values = wes_palette("AsteroidCity1")) +
  # adding labels to each dotted line from the `precipitation_peaks` object
  geom_text(data = precipitation_peaks,
            # x-axis location: date
            # y-axis location: set y value from earlier chunk
            # label = the defined label from earlier chunk
            # colro by label
            aes(x = date, y = label_y, label = label, color = label),
            # 45 degree angle for readability
            angle = 45,
            # slightly to right of vertical dotted line
            hjust = -0.1,
            # text sixe
            size = 3,
            # text bolding
            fontface = "bold")
```

```
Warning: Removed 3 rows containing missing values or values outside the scale range
(`geom_col()`).
```

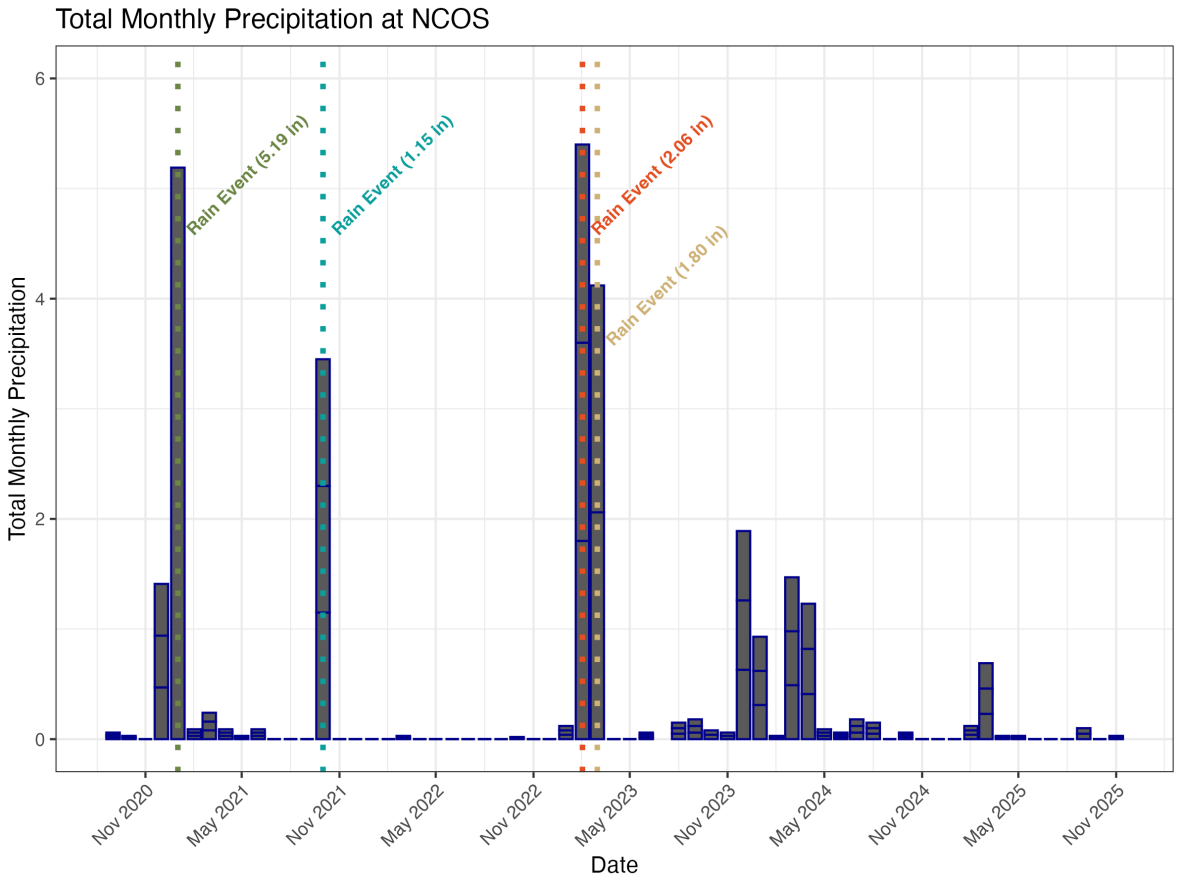


Figure 2. Salinity Time Series w/ Prcp Highlights

```
# base layer ggplot
# data: salinity_prcp
ggplot(data = salinity_prcp,
       # x-axis: date
       mapping = aes(x = date,
                     # y-axis: mean_salinity
                     y = mean_salinity)) +
  # column displaying mean salinity
  geom_col(color = "grey") +
  # separating points from sites into different plots
  facet_wrap(~ site_full) +
  # adding themeing
  theme_bw() +
```

```
  # x breaks by year to reduce cognitive load, labeling by abbreviated month and full year
  scale_x_date(date_breaks = "12 months", date_labels = "%b %Y") +
  # angling x axis labels to 45 degrees and to be right-aligned
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  # labeling x and y axes
  labs(x = "Date",
       y = "Monthly Average Salinity (ppt)",
       # adding title
       title = "Monthly Average NCOS Salinity",
       # adding legend title
       color = "Precipitation Events") +
  # adding lines from the `precipitation_peaks_facet` object
  geom_vline(data = precipitation_peaks_facet,
             # oriented on the date
             aes(xintercept = date,
                 # different colors by label
                 color = label),
             # long dash for better viz
             linetype = "longdash",
             # line width
             linewidth = 0.75) +
  # custom colors
  scale_color_manual(values = wes_palette("AsteroidCity1")) +
  # legend at bottom for less visual clutter
  theme(legend.position = "bottom")
```

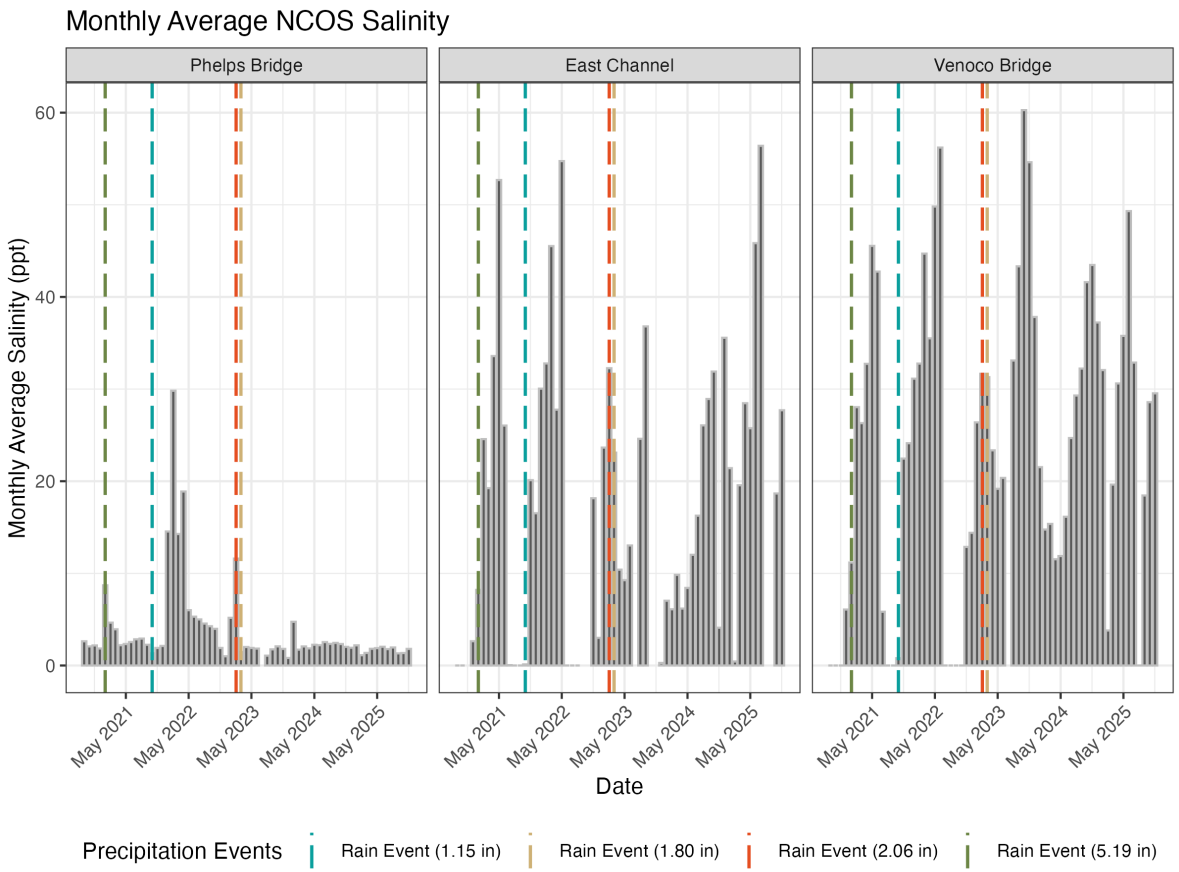


Figure 3. - How Salinity Relates to Water Surface Elevation (Venoco Bridge)

```
ggplot(data = elv_viz,
       mapping = aes(x = avg_elv,
                     y = mean_salinity)) +
  geom_point(color = "maroon") +
  facet_wrap(~ site_full) +
  scale_color_manual(values = wes_palette("GrandBudapest1")) +
  theme_bw() +
  labs(x = "Monthly Average Water Surface Elevation (m)",
       y = "Monthly Average Salinity (ppt)",
       title = "Figure 2. How Salinity Relates to Water Surface Elevation (Venoco Bridge)")
```

Figure 2. How Salinity Relates to Water Surface Elevation (Venoco Bridge)

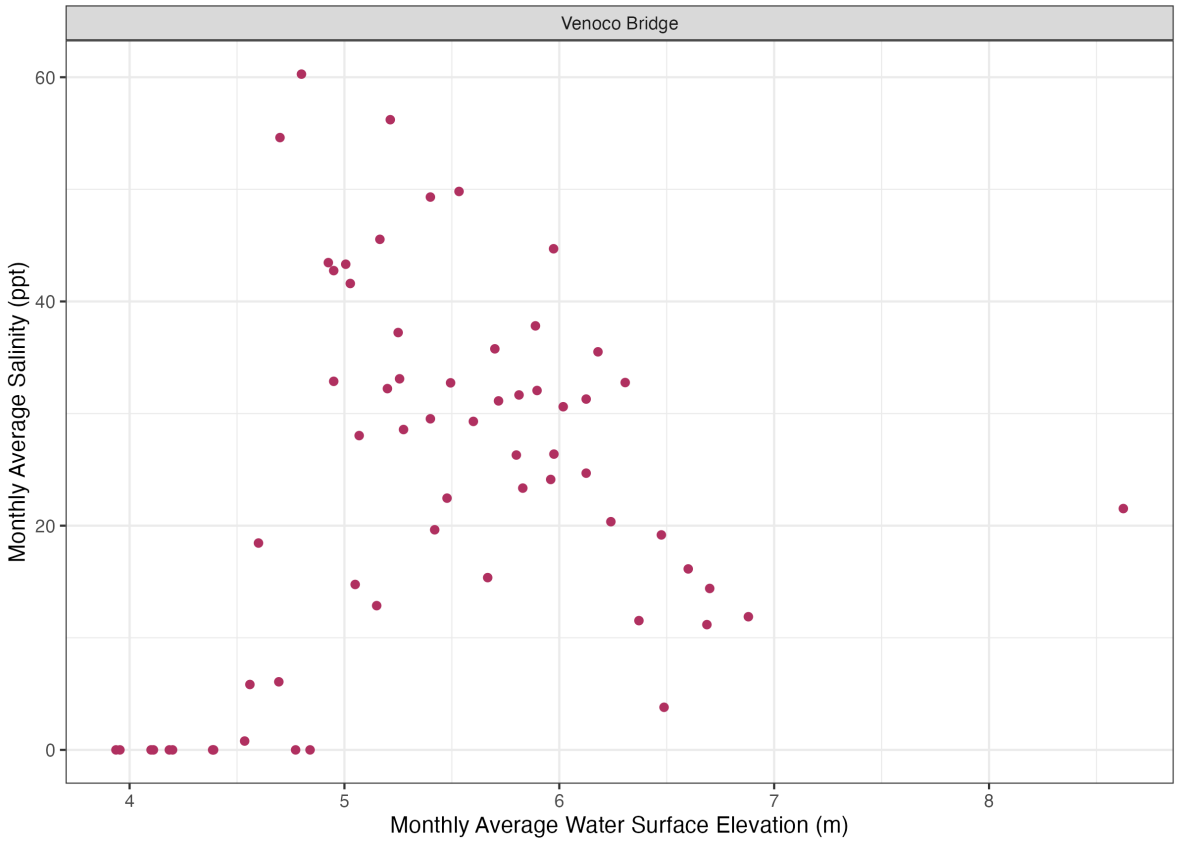


Figure 4. - How Salinity Varies Between Elevation Codes

```
ggplot(data = exploratory_salinity,
       mapping = aes(x = elevation_code,
                     y = mean_salinity)) +
  geom_point(shape = 21) +
  stat_summary(fun = median,
               geom = "point",
               color = "maroon") +
  stat_summary(fun = median,
               geom = "line",
               color = "maroon",
               group = 1) +
  labs(x = "Elevation Code",
       y = "Mean Salinity",
```

```
       title = "Figure 3. Mean Salinity Differences Between Elevation Codes") +
  facet_wrap(~ site_full) +
  theme_bw() +
  scale_x_discrete(breaks = c(0,1, 2, 3)) +
  scale_y_continuous(position = "right") +
  coord_flip()
```

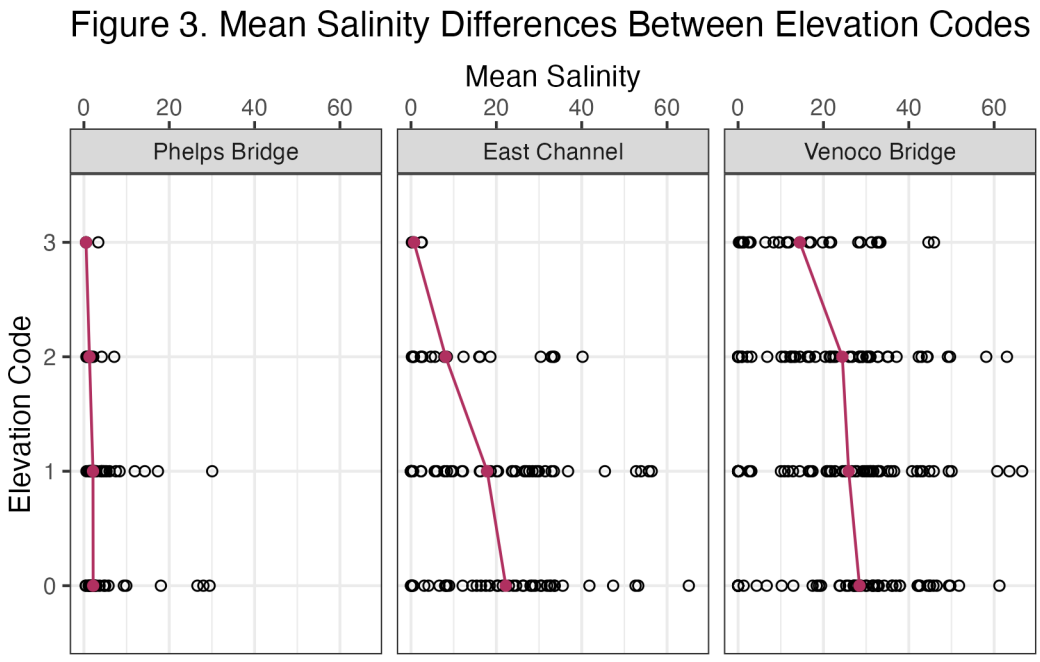
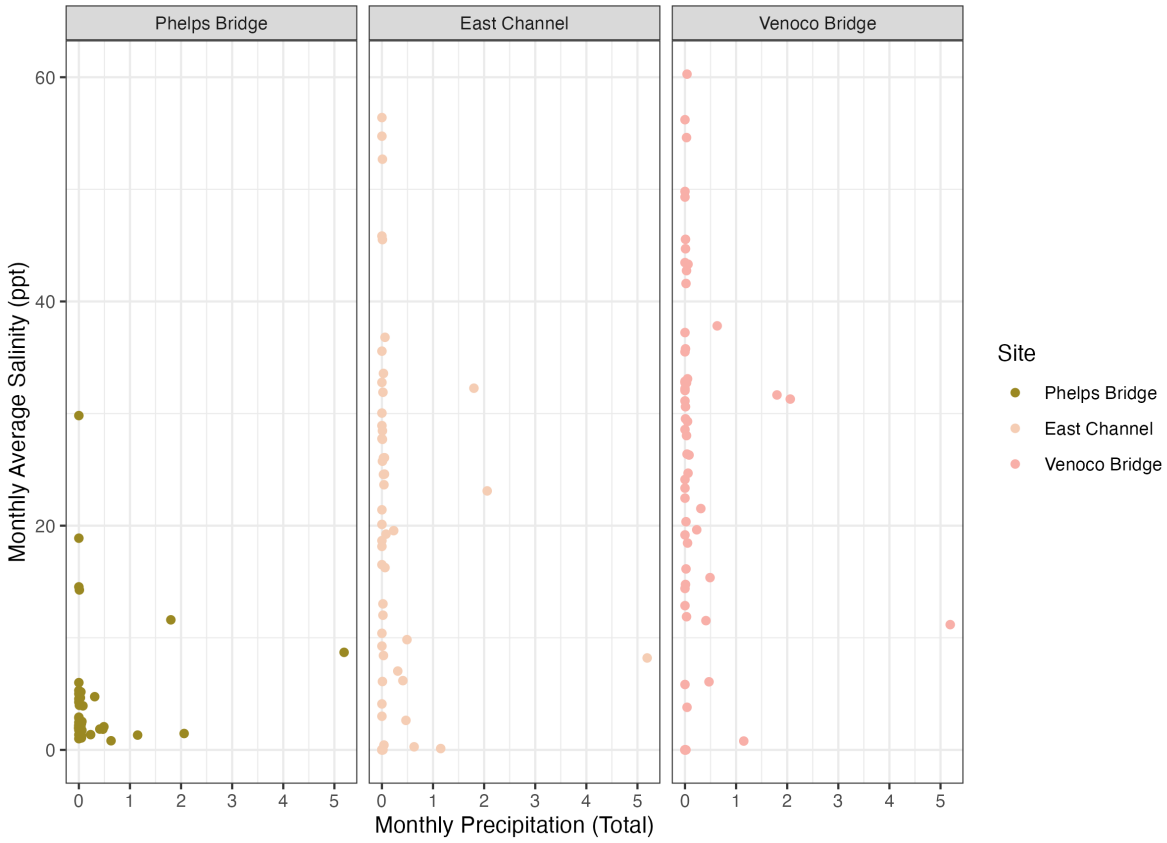


Figure 5. - How Precipitation Explains Salinity

```
ggplot(data = salinity_prcp,
       mapping = aes(x = sum_prcp,
                     y = mean_salinity,
                     color = site_full)) +
  geom_point() +
  facet_wrap(~ site_full) +
  scale_color_manual(values = wes_palette("Royal2")) +
  theme_bw() +
  labs(x = "Monthly Precipitation (Total)",
       y = "Monthly Average Salinity (ppt)",
       title = "Figure 4. How Precipitation Explains Salinity",
       color = "Site")
```

Figure 4. How Precipitation Explains Salinity



Correlation Test

Calculating Spearman's Correlation

```
cor.test(salinity_prcp$mean_salinity,
         salinity_prcp$sum_prcp,
         method = "spearman")
```

```
Warning in cor.test.default(salinity_prcp$mean_salinity,
salinity_prcp$sum_prcp, : Cannot compute exact p-value with ties
```

```
    Spearman's rank correlation rho
```

```
data:  salinity_prcp$mean_salinity and salinity_prcp$sum_prcp
S = 939961, p-value = 0.8251
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
0.01663311
```

Interpreting the Results

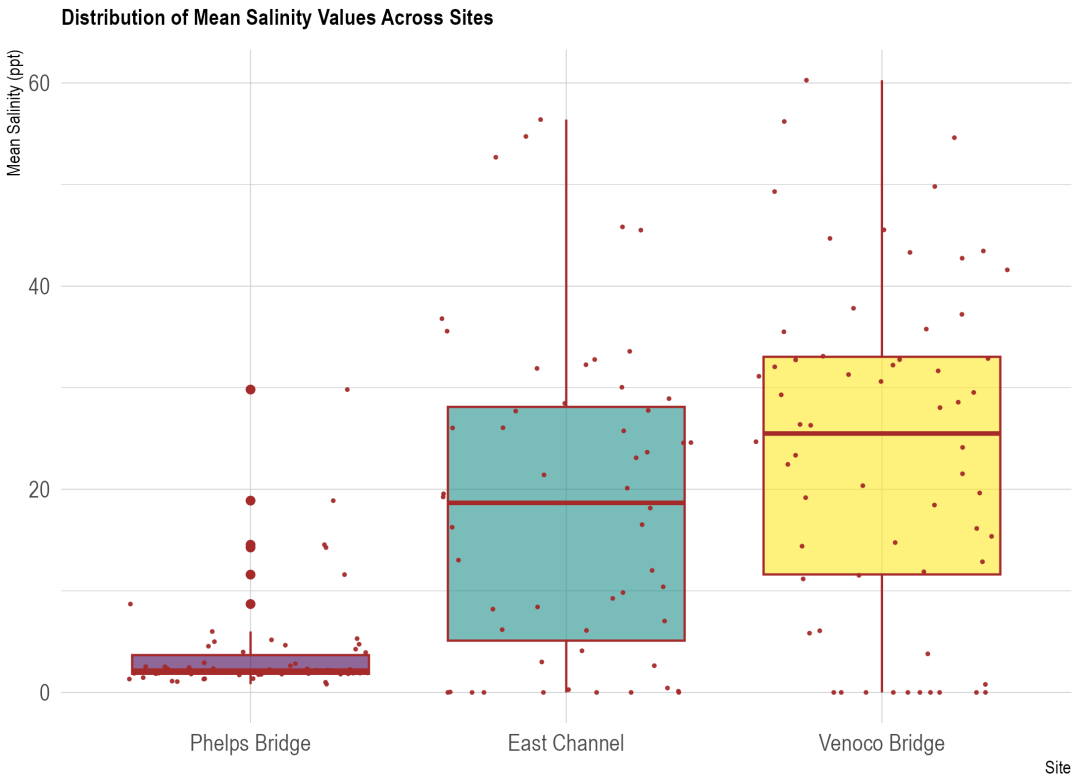
A Spearman's rank correlation was conducted to examine the relationship between monthly precipitation and monthly average salinity. Results indicated no statistically significant relationship between the two variables (rho = 0.017, S = 939,961, p = 0.825). The near-zero rho suggests that monthly precipitation has negligible association with average salinity at this scale. It should be noted that exact p-values could not be computed due to ties in the data, so a normal approximation was used; this does not meaningfully affect the interpretation. These results may reflect lag effects between precipitation events and salinity response, spatial or hydrological disconnect between measurement sites, or the influence of other drivers, such as evaporation or tidal exchange, that operate independently of local precipitation.

4. Other exploratory visualizations

Comparing Across Groups

Boxplot w/ Jitter - Distribution of Mean Salinity Values Across Sites

```r
# Plot
salinity_prcp |>
  ggplot( aes(x= site_full, y=mean_salinity, fill = site_full)) +
    geom_boxplot(color = "brown") +
    scale_fill_viridis(discrete = TRUE, alpha=0.6) +
    geom_jitter(color="brown", size=0.4, alpha=0.9) +
    theme_ipsum() +
    theme(
      legend.position="none",
      plot.title = element_text(size=11)
    ) +
    ggtitle("Distribution of Mean Salinity Values Across Sites") +
    xlab("Site") +
  ylab("Mean Salinity (ppt)")
```

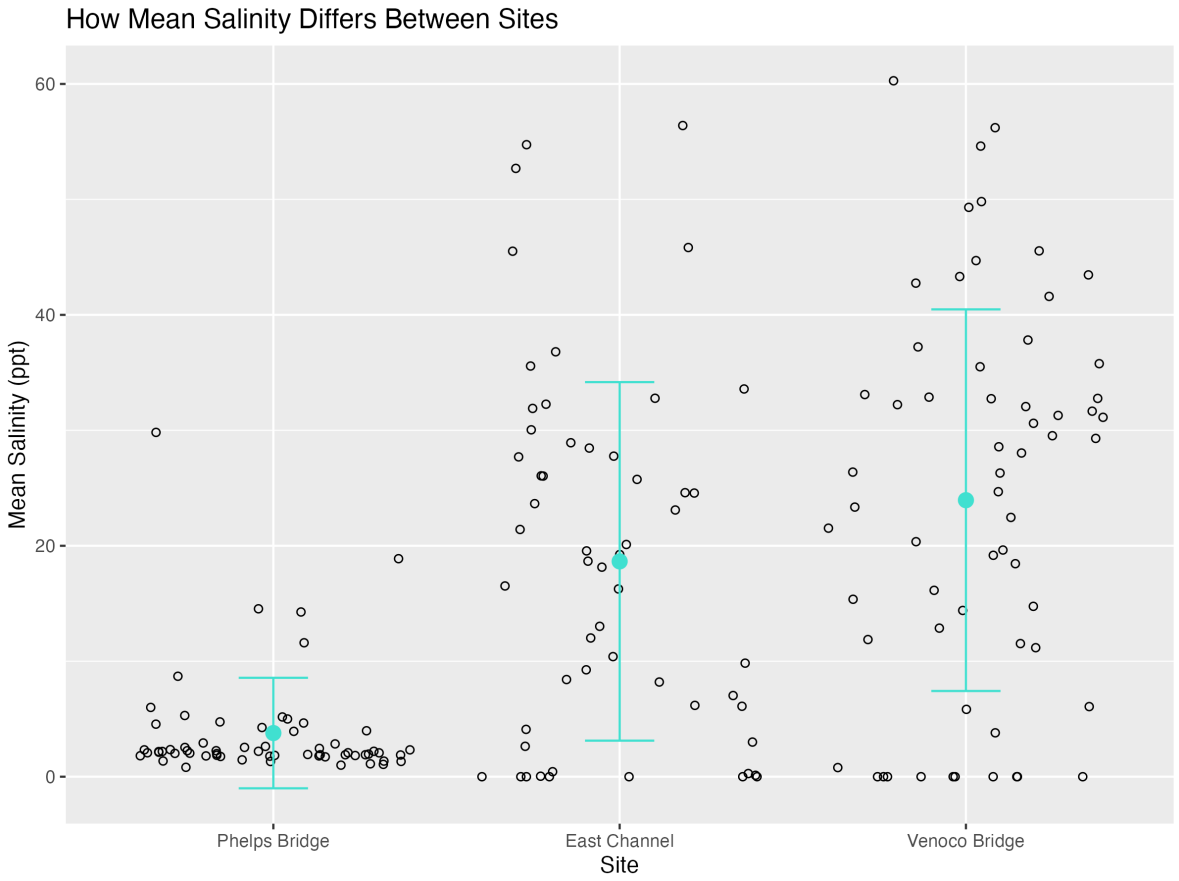


- The addition of this plot was taken as a suggestion made by An after reviewing our Work Plan/Project Proposal. This figure aims to demonstrate that not only dooes the elevation code of a site yield a difference in monthly mean salinity (as seen in figure 3), but also that the three NCOS sites also differ in their overall monthly mean salinity values. This plot sets the basic understanding of salinity variety across sites while also adding context to one of the visualizations directly answering our question.

Strip Chart w/ Standard Deviation - How Mean Salinity Differs Between Sites

```
ggplot(data = salinity_prcp,
       mapping = aes(x = site_full,
                     y = mean_salinity)) +
  geom_jitter(shape = 21) +
  stat_summary(fun = mean, geom = "point", size = 3, color = "turquoise") +
  stat_summary(fun.data = mean_sdl, fun.args = list(mult = 1), geom = "errorbar", width = 0.
```

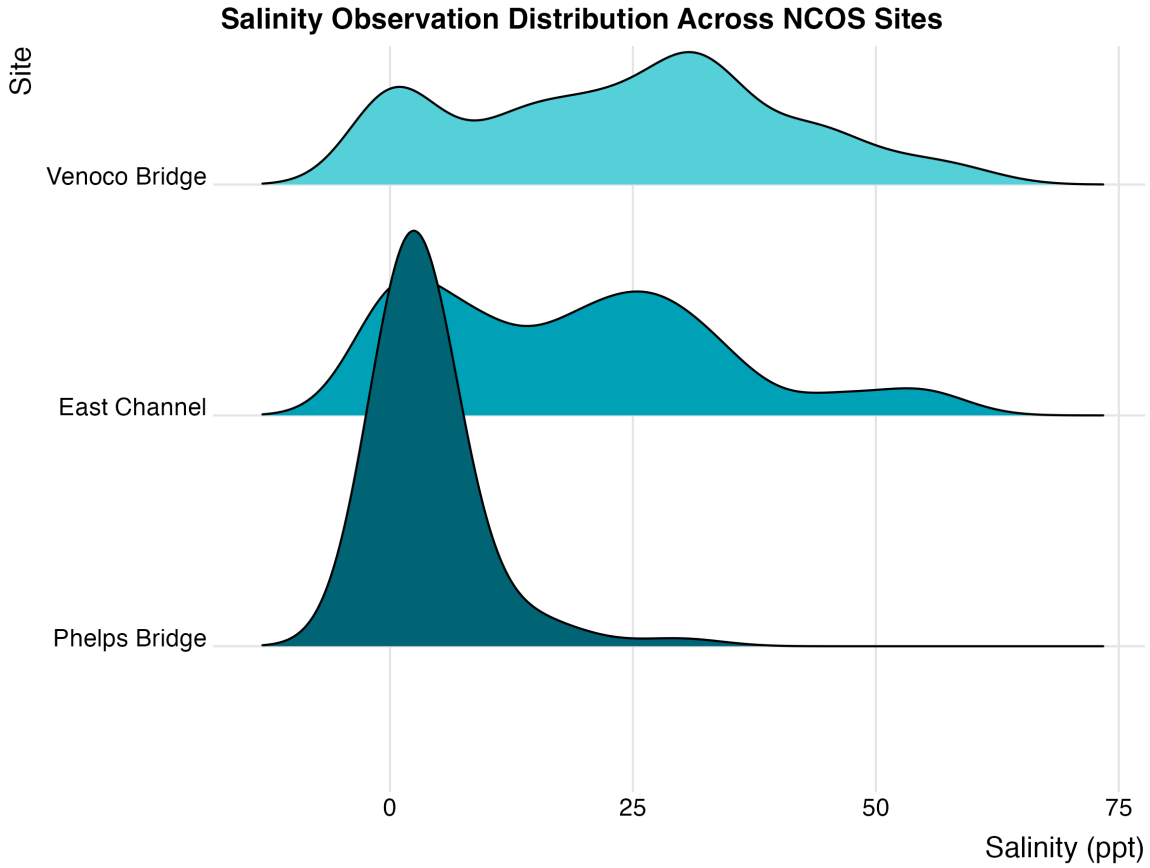
```
  labs(x = "Site",
       y = "Mean Salinity (ppt)",
       title = "How Mean Salinity Differs Between Sites")
```



Ridgeline Plot - Salinity Observation Distribution Across NCOS Sites

```
ggplot(salinity_prcp, aes(x = mean_salinity, y = site_full, fill = site_full)) +
  geom_density_ridges() +
  theme_ridges() +
  theme(legend.position = "none") +
  scale_fill_manual(values = natparks.pals("Banff")) +
  labs(x = "Salinity (ppt)",
       y = "Site",
       title= "Salinity Observation Distribution Across NCOS Sites")
```

```
Picking joint bandwidth of 4.37
```

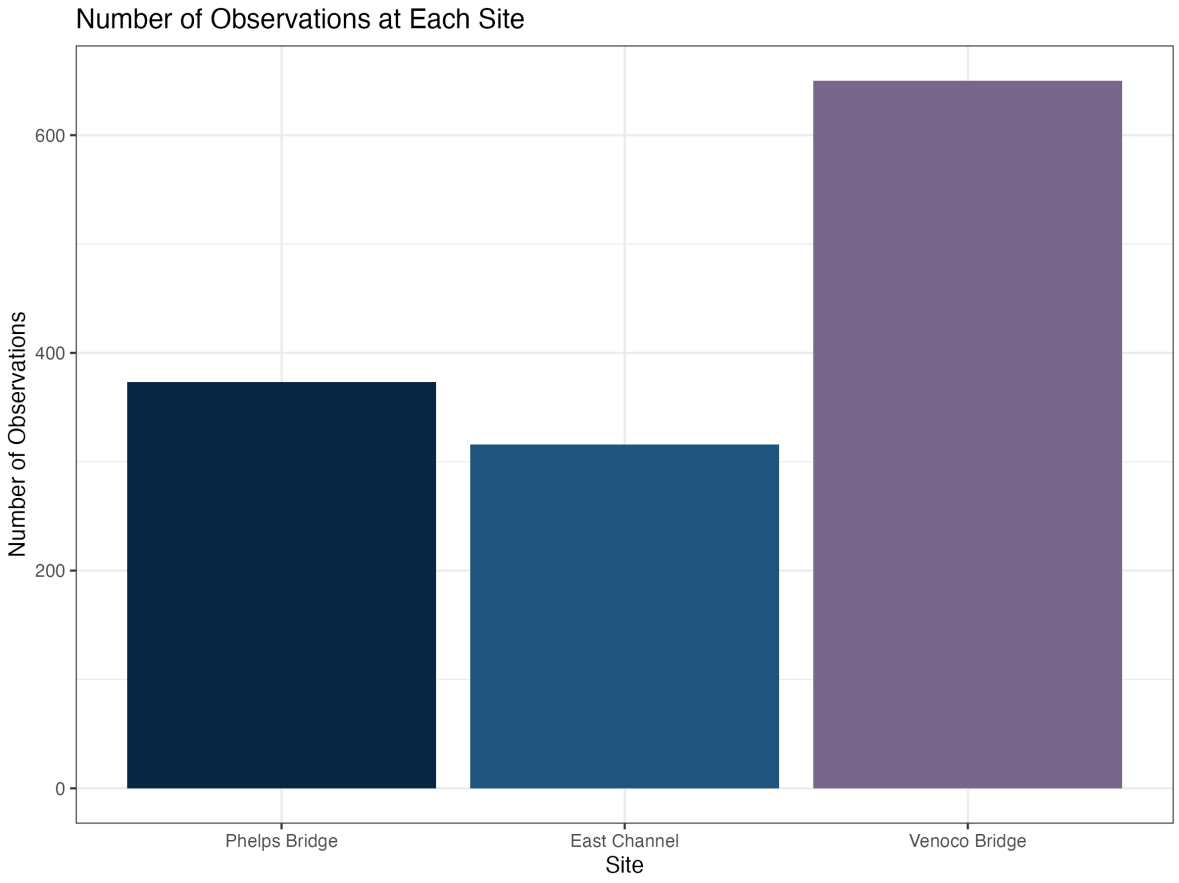


Distribution of Observations

of Observations per Site

```
ggplot(data = water_quality_weather |>
         filter(!global_id %in% salinity_outlier_ids,
                site_name != "other",
                elevation_code %in% c("0", "1", "2")) |>
         mutate(site_full = fct_relevel(site_full,
                                        "Phelps Bridge", "East Channel", "Venoco Bridge")),
       mapping = aes(x = site_full)) +
  scale_fill_manual(values = natparks.pals("Volcanoes")) +
  geom_bar(aes(fill = site_full)) +
```

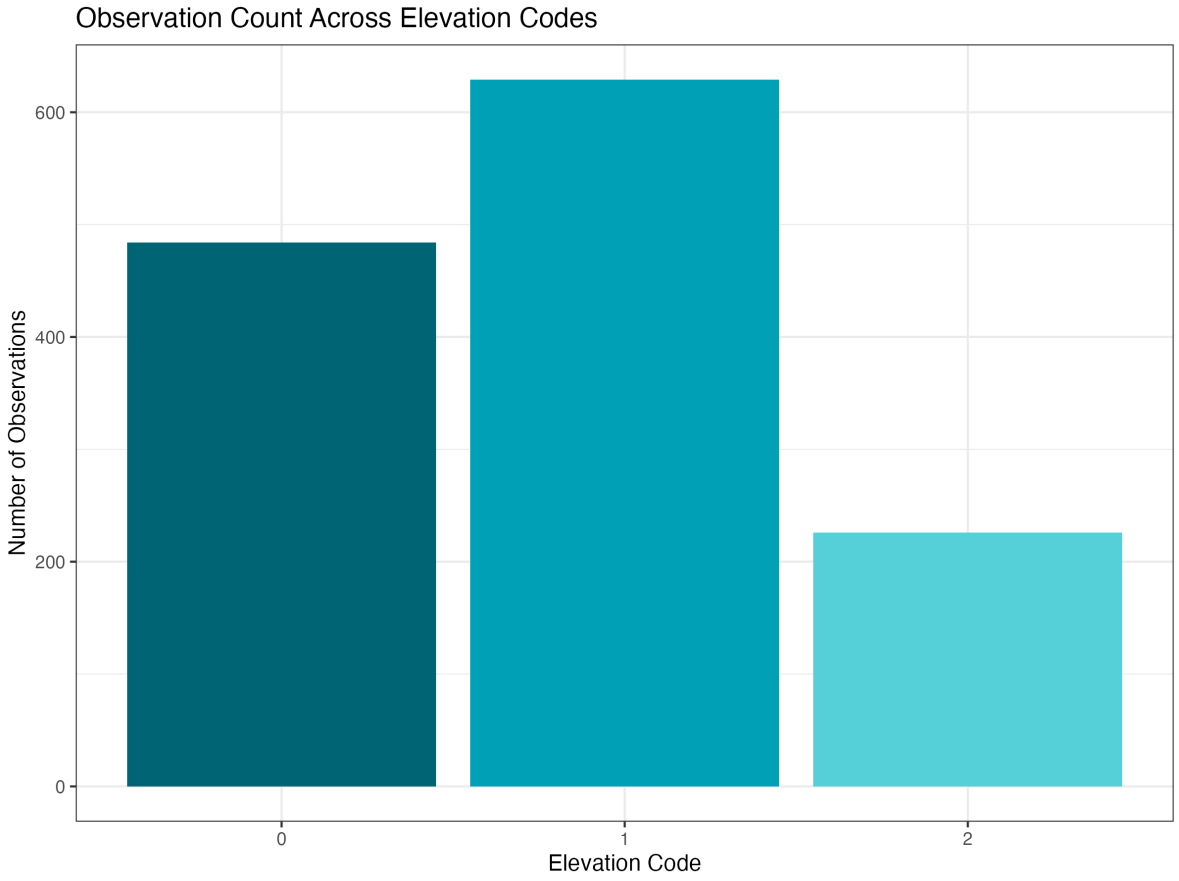
```
  theme_bw() +
  theme(legend.position = "none") +
  labs(x = "Site",
       y = "Number of Observations",
       title = "Number of Observations at Each Site")
```



of Observations per Elevation Code

```
ggplot(data = water_quality_weather |>
         filter(!global_id %in% salinity_outlier_ids,
                site_name != "other",
                elevation_code %in% c("0", "1", "2")),
       mapping = aes(x = elevation_code)) +
  geom_bar(aes(fill = elevation_code)) +
```

```
  scale_fill_manual(values = natparks.pals("Banff")) +
  theme_bw() +
  theme(legend.position = "none") +
  labs(x = "Elevation Code",
       y = "Number of Observations",
       title = "Observation Count Across Elevation Codes")
```



Both of these figures were added last-minute to the work plan after our data visualization presentation in class. We feel that the addition of this informtion is important in contextualizing our outputs. Venoco Bridge has the most amount of observations, followed by Phelps Bridge, and finally, East Channel. We can also see that elevation code 1 has the most amount of observations, followed by 0, anf finally, 2. Understanding the distribution of observations allows for the appropriate selection of statistical methods, correct interpretation of analytical results, and effective and accurate communication of findings.

\\

5. Plan for Elective